

Solving Equations & Inequalities

Marking Scheme & Blueprint

MTH1W · Module 3 Test — Teacher Copy

Blueprint

Curriculum: MTH1W (Grade 9 de-streamed math). Total: 70 marks · 60 minutes · 30 questions. Categories: K/U, Application (App), Thinking (Think), Communication (Comm).

Q	Sub-topic	Level	Marks	Category	Section
A1	One-Step	L1	1	K/U	A
A2	One-Step	L1	1	K/U	A
A3	One-Step	L1	1	K/U	A
A4	One-Step	L1	1	K/U	A
A5	Two-Step	L1	1	K/U	A
A6	Two-Step	L1	1	K/U	A
A7	Inequalities	L1	1	K/U	A
A8	One-Step	L1	1	K/U	A
A9	One-Step	L1	1	K/U	A
A10	Two-Step	L1	2	K/U	A
A11	Two-Step	L1	2	K/U	A
A12	Fractions	L1	2	K/U	A
A13	Distribution	L1	2	K/U	A
A14	Both Sides	L1	2	K/U	A
A15	Inequalities	L1	2	K/U	A
B16	Both Sides	L2	2	App	B
B17	Fractions	L2	2	App	B
B18	Distribution	L2	2	App	B
B19	Rearranging	L2	2	App	B
B20	Two-Step	L2	2	App	B
B21	Inequalities	L2	2	App	B
B22	Distribution	L2	3	App	B
B23	Rearranging	L2	3	App	B
B24	Fractions	L2	3	App	B
C25	Consecutive Integers	L3	4	Think	C
C26	Inequalities (flip)	L3	4	Think	C
C27	Formula & Evaluate	L3	5	Think	C
C28	Fractions	L3	5	Think	C
C29	Word Problem	L3	6	Think	C
D30	Reasoning on Inequalities	L3	4	Comm	D

By category (marks): K/U 21 · Application 21 · Thinking 24 · Communication 4

By level (marks): L1 21 · L2 21 · L3 28 → L1 30% · L2 30% · L3 40%

Worked Solutions & Mark Allocation

- 1. (A1 · K/U · 1 marks)**
 $x = 7$. (A1)
- 2. (A2 · K/U · 1 marks)**
 $x = 13$. (A1)
- 3. (A3 · K/U · 1 marks)**
 $x = 8$. (A1)
- 4. (A4 · K/U · 1 marks)**
 $x = 18$. (A1)
- 5. (A5 · K/U · 1 marks)**
 $2x = 10$, $x = 5$. (A1)
- 6. (A6 · K/U · 1 marks)**
 $5x = 20$, $x = 4$. (A1)
- 7. (A7 · K/U · 1 marks)**
 $x > 4$. (A1)
- 8. (A8 · K/U · 1 marks)**
 $y = 3$. (A1)
- 9. (A9 · K/U · 1 marks)**
 $x = 7$. (A1)
- 10. (A10 · K/U · 2 marks)**
 $4x = 24$, $x = 6$. (M1, A1)
- 11. (A11 · K/U · 2 marks)**
 $5x = 25$, $x = 5$. (M1, A1)
- 12. (A12 · K/U · 2 marks)**
 $x/4 = 3$, $x = 12$. (M1, A1)
- 13. (A13 · K/U · 2 marks)**
 $2x + 10 = 16$, $2x = 6$, $x = 3$. (M1, A1)
- 14. (A14 · K/U · 2 marks)**
 $4x = 12$, $x = 3$. (M1, A1)
- 15. (A15 · K/U · 2 marks)**
 $x \leq 5$; closed circle at 5, arrow left. (M1, A1)
- 16. (B16 · Application · 2 marks)**
 $6x - 2x = 15 + 5$, $4x = 20$, $x = 5$. (M1, A1)

- 17. (B17 • Application • 2 marks)**
 $2x/3 = 6$, $2x = 18$, $x = 9$. (M1, A1)
- 18. (B18 • Application • 2 marks)**
 $3x - 12 = 2x + 1$, $x = 13$. (M1, A1)
- 19. (B19 • Application • 2 marks)**
 Divide both sides by lw : $h = V/(lw)$. (M1, A1)
- 20. (B20 • Application • 2 marks)**
 $-3x = -12$, $x = 4$. (M1, A1)
- 21. (B21 • Application • 2 marks)**
 $3x > 9$, $x > 3$; open circle at 3, arrow right. (M1, A1)
- 22. (B22 • Application • 3 marks)**
 $4x + 8 = 2x + 18$. (M1)
 $2x = 10$, $x = 5$. (M1, A1)
- 23. (B23 • Application • 3 marks)**
 Multiply by 2: $2A = bh$. (M1)
 Divide by h : $b = 2A/h$. (M1, A1)
- 24. (B24 • Application • 3 marks)**
 Multiply every term by 6: $3x + 2x = 30$. (M1)
 $5x = 30$, $x = 6$. (M1, A1)
- 25. (C25 • Thinking • 4 marks)**
 Let them be n , $n+1$, $n+2$: $n + (n+1) + (n+2) = 51$. (M1)
 $3n + 3 = 51$, $3n = 48$. (M1)
 $n = 16$, so the integers are 16, 17, 18. (A1, A1)
- 26. (C26 • Thinking • 4 marks)**
 $-4x \geq 16$. (M1)
 Divide by -4 and reverse: $x \leq -4$. (M1, A1)
 Graph: closed circle at -4 , arrow left. (A1)
- 27. (C27 • Thinking • 5 marks)**
 $F - 32 = (9/5)C$. (M1)
 $C = (5/9)(F - 32)$. (M1, A1)
 At $F = 212$: $C = (5/9)(180) = 100$. (M1, A1)
- 28. (C28 • Thinking • 5 marks)**
 Multiply every term by 12: $4(x + 2) - 3(x - 1) = 24$. (M1)
 $4x + 8 - 3x + 3 = 24$, $x + 11 = 24$. (M1, M1)
 $x = 13$. (A1, A1)

29. (C29 • Thinking • 6 marks)

Let width = w , length = $2w + 3$. (M1)

$P = 2(2w + 3) + 2w = 6w + 6 = 48$. (M1, M1)

$6w = 42$, $w = 7$. (M1, A1)

Length = $2(7) + 3 = 17$ cm. (A1)

30. (D30 • Communication • 4 marks)

Dividing (or multiplying) both sides by a negative number reverses the direction of the inequality. (C1)

Divide by -2 and reverse: $x < -3$. (A1, C1)

Graph: open circle at -3 , arrow left. (A1)